

# Legal Postures for Search and Optimization in Redistricting

Gio Della-Libera

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## Abstract

This draft explains how search and optimization choices can support or weaken legal and policy narratives. It is a legal-interpretation paper, not a claim that algorithmic search is neutral, court-approved, or sufficient by itself. It uses "posture" to mean an evidence-backed argument position, not a legal conclusion.

## 1 Introduction

U.15 is a legal-interpretation paper. It describes how search, ensemble, optimization, and certification workflows affect legal posture without treating algorithmic output as inherently neutral.

The paper is intentionally bounded. It does not advise a jurisdiction, predict case outcomes, or translate audit pass/fail into compliance. Instead, it asks what kind of record an algorithmic workflow leaves behind and how that record can strengthen, weaken, or fail to address a stated policy or legal narrative.

## 2 Legal Postures

Search can support posture claims such as transparency, robustness, least-change comparison, alternative-plan availability, or auditability. Each posture requires different evidence and should be tied to specific commands, artifacts, and limitations.

The posture vocabulary is useful because it keeps computational artifacts in their lane. A search workflow can show that many alternatives were considered, or that a selected plan survived a declared audit profile. It cannot decide which legal standard applies, which evidence a court or agency will credit, or whether unmodeled facts change the conclusion.

## 3 Audit Artifacts

RPLAN/RCTX artifacts can show what was generated, selected, verified, and reported. They cannot by themselves establish constitutional compliance, statutory adequacy, or the absence of discriminatory effect.

| Posture                  | Supporting record                             | Common overclaim                        |
|--------------------------|-----------------------------------------------|-----------------------------------------|
| Transparency             | Parameters, seeds, constraints, and RCTX      | The process is neutral                  |
| Robustness               | Alternative plans under declared protocols    | The chosen plan is legally safe         |
| Least change             | Baseline, distance metric, and comparison set | The change is justified by metric alone |
| Alternative availability | Generated candidate set and filters           | Any alternative is equally viable       |
| Auditability             | RPLAN/RCTX, certificates, and manifests       | Audit pass proves compliance            |

Table 1: Legal/policy postures and the records they require.

For legal-facing use, the audit package should be paired with a claim memo that names the posture, evidence level, artifact set, omitted evidence, and jurisdiction-specific assumptions. The algorithmic record is an input to that memo, not the memo itself.

## 4 Current Implementation Boundary

The legal posture of a search result depends on the artifact class. A seed ledger supports reproducibility; an ensemble summary supports a comparison under declared diagnostics; a solver certificate supports a model-scoped bound; and an RPLAN/RCTX audit package supports verifier replay of declared plan properties. None of these artifacts alone proves statewide legal sufficiency.

The current U.16–U.20 papers provide the stronger implementation reference points. Older posture examples should be read as argument templates that must be backed by those package and certificate forms before they are used in commission or court-facing settings.

## 5 Limitations

This paper does not offer legal advice or claim that any workflow guarantees compliance. It separates computational evidence from legal interpretation and marks jurisdiction-specific questions as outside the algorithmic certificate.

The draft is also incomplete as legal scholarship until it is paired with jurisdiction-specific sources and concrete command transcripts. Its immediate role is portfolio discipline: every U-series paper should state whether its output supports transparency, robustness, comparison, auditability, or some other posture, and should name what remains outside the record.

## 6 Conclusion

U.15 gives the U-series its legal claim discipline: search and optimization can strengthen transparency, but legal posture depends on evidence, context, and explicitly bounded inter-

pretation.

That discipline is what makes the technical papers more usable. They can produce plans, traces, certificates, and manifests while leaving the final legal interpretation to a separately stated posture with its own assumptions and limits.

## References

BISECT Project. U.15 legal postures for search plan, 2026. BISECT internal paper plan.